HR ANALYTICS EMPLOYEES ATTRITION

FOUR WEEKS INDUSTRIAL TRAINING PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF
THE REQUIREMENT FOR THE AWARD OF
DEGREE OF

BACHELOR OF TECHNOLOGY

(COMPUTER SCIENCE & ENGINEERING)

SUMBMITTED BY

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Certificate



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CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the project entitled **"HR Analytics Employee Attrition using Python"** by **Rupam Kaur**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering**, submitted to **Sant Baba Bhag Singh University**, is an authentic record of my own work carried out during the period of my industrial training.

The work presented in this project has been completed by me during my training period under the guidance of the concerned training supervisor. The matter presented in this project has not been submitted by me in any other University or Institute for the award of any degree.

Signature of the Student: RUPAM KAUR

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Date: _____

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Sant Baba Bhag Singh University** for providing me with the opportunity to undertake this industrial training and complete my project entitled **"HR Analytics Employee Attrition using Python."**

The successful completion of my training and project has been possible through the guidance and encouragement of my mentors. I sincerely appreciate my training instructors and project guide for sharing their expertise and helping me gain practical exposure to Python programming, data analysis, visualization techniques, and Streamlit application development.

This project has also been a valuable learning experience, and I would like to acknowledge my teachers and department for creating a supportive academic environment. Their guidance helped me strengthen my technical knowledge and apply the skills acquired during my training.

I am deeply thankful to my family and friends for their patience, motivation, and encouragement throughout this journey. I also appreciate the efforts of all those who, in any way, contributed to my learning experience and the completion of this project.

ABSTRACT

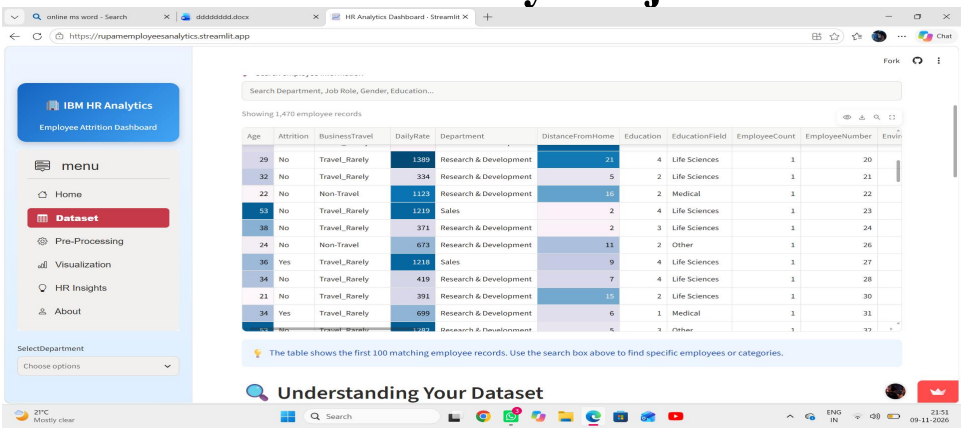
The HR Analytics Employee Attrition project focuses on analyzing employee data to understand the factors related to employees leaving an organization. The main objective of this project is to study employee information, identify useful patterns, and present meaningful insights through an interactive dashboard.

The project uses the IBM HR Analytics Employee Attrition dataset, which contains 1,470 rows and 35 columns. Each row represents an employee, while the columns include information such as Age, Department, Job Role, Monthly Income, Job Satisfaction, Work Experience, and Attrition.
The Data Preprocessing stage involves examining the dataset, checking its structure, and identifying missing values and duplicate records. The dataset contains no missing values and no duplicate records. Therefore, no missing-value filling or duplicate-row removal was required. The data was examined and prepared for further analysis.
Exploratory Data Analysis (EDA) was performed using Python libraries such as Pandas, NumPy, Matplotlib, Plotly, and Seaborn. Different charts and graphs were used to explore employee characteristics, compare employee categories, and understand attrition-related patterns.

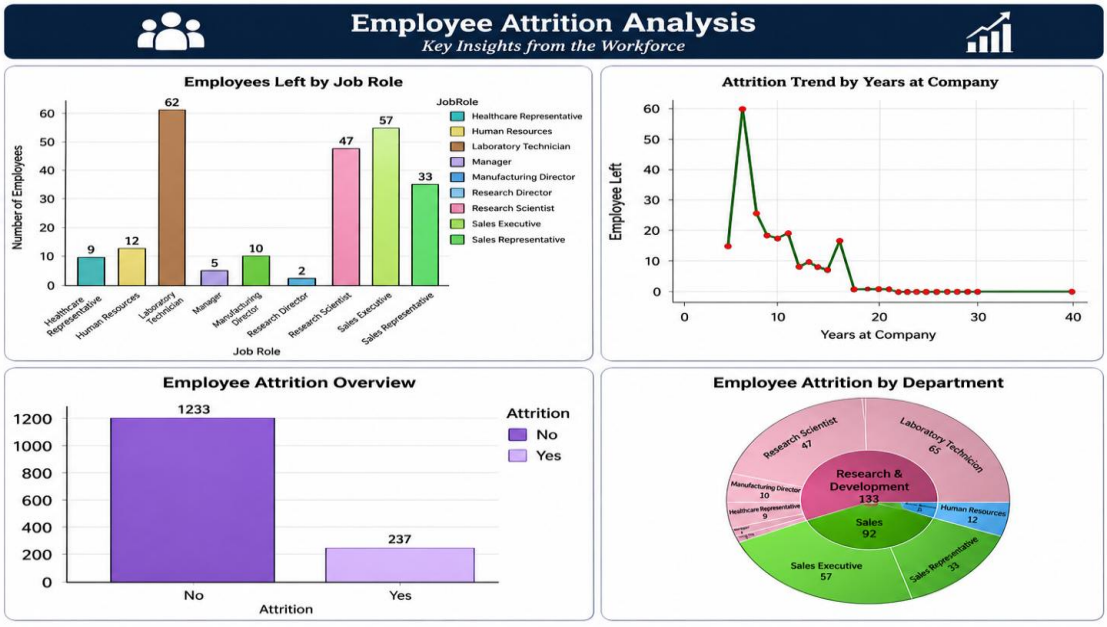
The project is presented through an interactive Streamlit Dashboard consisting of six sections: Home, Dataset, Preprocessing, Visualization, HR Insights, and About. The Home section introduces the project, Dataset provides employee data details, and Preprocessing explains the data preparation process.

The Visualization section presents charts and graphs, while HR Insights highlights important findings.
The About section provides information about the project and its technologies.
Overall, this project demonstrates the practical use of Data Analysis and Data Visualization in Human Resource Analytics. It helps users explore employee data, understand attrition-related trends, and shows how an interactive dashboard can support better decision-making in Human Resource Management.

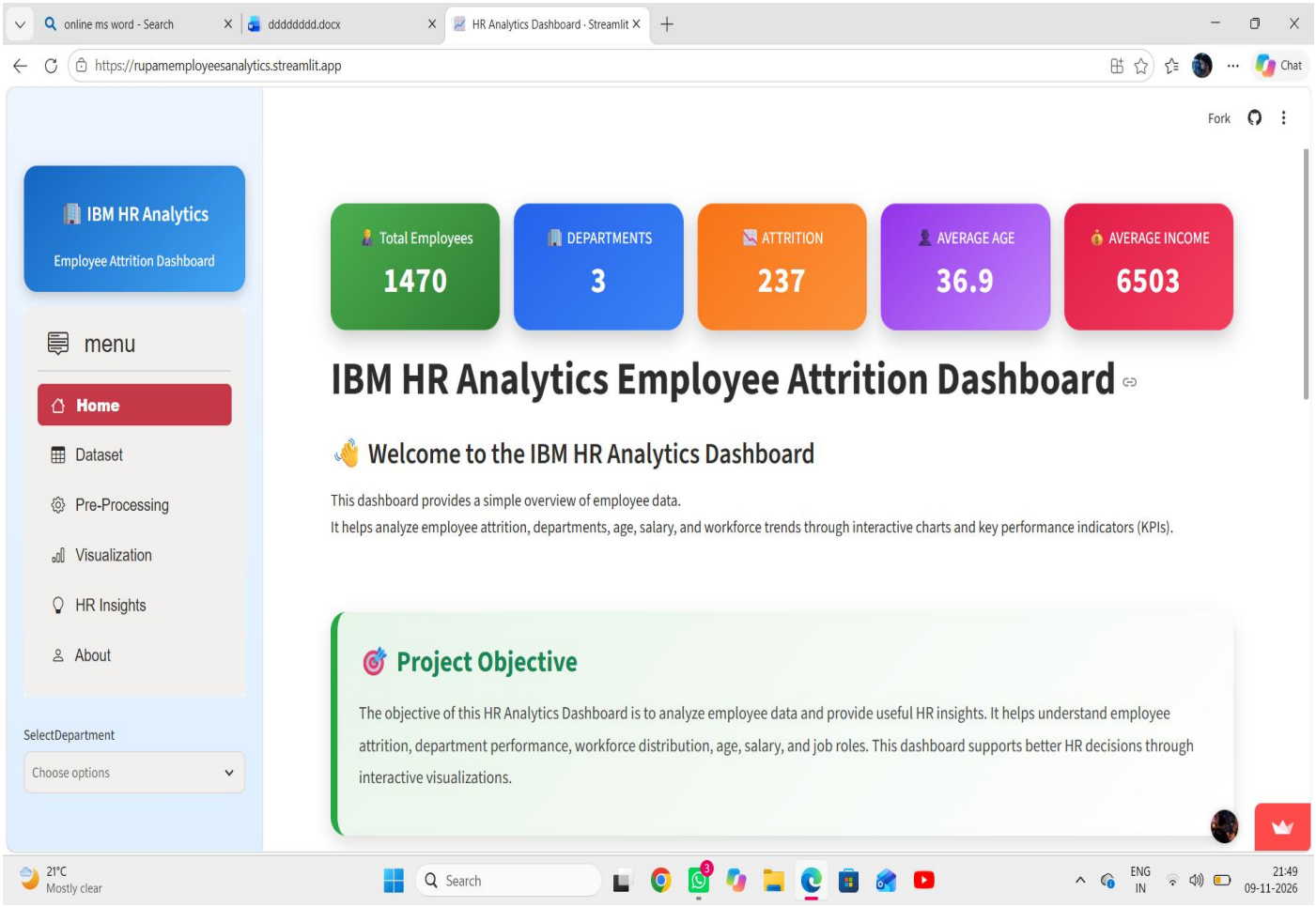
Dataset of my Project



Insights of my Project



- # Home Page of my Project.



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ABOUT THE COMPANY / INDUSTRY

Q7 Services is the organization where the industrial training was completed. The training was based on **Python Programming with Data Science**, with practical learning of programming, data handling, data preprocessing, data analysis, and data visualization.

During the training, practical work was performed on an **HR Analytic Employee Attrition.** The training provided practical experience in cleaning and analyzing data and presenting the results through visualizations. The knowledge gained during the training was applied to develop the project **"HR Analytic Employee Attrition"** and its interactive **Streamlit dashboard**.

The training helped in understanding how Python and data science concepts can be applied to a real-world dataset and strengthened practical skills in data analysis and visualization.

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LIST OF FIGURES

FIGURE 4.1	**HOME OF HRANALYTIC EMPLOYEE ATTRITION**
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FIGURE 7.3	**HR INSIGHTS OF PAGE**
FIGURE 7.4	**VISUALIZATION OF PAGE**
FIGURE 7.5	**ABOUT PAGE**

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CHAPTER 1

INTRODUCTION TO ORGANIZATION

1.1 Industry Profile

Data Science is one of the rapidly growing fields in the Information Technology industry. It involves collecting, processing, analyzing, and interpreting data to extract meaningful information. Organizations generate large amounts of data through their daily activities, and Data Science helps them understand this data and make better decisions. It combines programming, statistics, mathematics, and data visualization to solve real-world problems.

Python Programming is widely used in Data Science because of its simple syntax and powerful libraries. It is useful for data processing, analysis, automation, and visualization. Libraries such as Pandas and NumPy help in handling datasets and performing numerical operations efficiently.

Data Analytics is an important part of Data Science. It involves examining data to identify trends, patterns, and relationships. Data analysis helps organizations understand their performance, solve problems, and make informed decisions based on available information.

The Human Resource Management industry is one area where data analysis can be useful because a large amount of information is generated regarding employees, departments, job roles, salaries, job satisfaction, work-life balance, and employee attrition. Analysis of such information can help in understanding the characteristics and patterns present in employee data.

The project developed during the training, "Human Resource Analytics," applies Data Science techniques to an employee dataset. The project includes data preprocessing, data analysis, visualization, and development of an interactive Streamlit dashboard.

1.2 General Information

The industrial training was focused on Python Programming with Data Science and provided practical exposure to the application of programming, data processing, and data analysis concepts. The training helped in understanding how Python can be used to handle real-world datasets and extract meaningful information. During the training, the project work was carried out on Human Resource Analytics, focusing on employee attrition using the IBM HR Analytics Employee Attrition dataset.

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The project follows a structured process in which the raw dataset is first examined to understand its structure, columns, and data types. The dataset contains 1,470 rows and 35 columns, including employee-related information such as age, department, job role, monthly income, job satisfaction, work-life balance, overtime, and years of experience. The data is checked for missing values and duplicate records to ensure that it is suitable for analysis.

After examining and preprocessing the dataset, different data analysis techniques are applied to understand employee-related patterns. Python libraries such as Pandas and NumPy are used for data handling and processing. The analysis focuses on employee attrition and helps in understanding how different employee-related factors can be studied through data.

The project also involves data visualization to present the analytical results in a clear and understandable manner. Libraries such as Matplotlib, Seaborn, and Plotly are used to represent data through charts and graphs. These visualizations make it easier to understand the patterns present in the dataset and provide a clear view of the analytical findings.

Furthermore, an interactive Streamlit dashboard is developed to present the project in an organized and user-friendly manner. The dashboard includes sections such as Home, Dataset, Preprocessing, Visualization, HR Insights, and About. These sections allow users to explore the dataset, understand the preprocessing process, view visualizations, and examine the information obtained through analysis.

The practical work involved understanding the dataset, preprocessing the data, performing analysis, and presenting the results using visualizations. The training therefore provided practical experience in applying Python Programming and Data Science concepts to a real-world employee dataset. The overall purpose of the project is to transform employee data into meaningful and understandable information through data preprocessing, analysis, visualization, and interactive dashboard development..

CHAPTER 2

INTRODUCTION TO PROJECT

2.1 Project Introduction

Human Resource Analytics is an application of Data Science that helps organizations analyze employee-related data and understand workforce trends. Employee attrition is an important area of HR Analytics, as it helps in studying patterns related to employees leaving an organization.

The project "Human Resource Analytics Employee Attrition using Python" focuses on analyzing employee data using Python Programming and Data Science techniques. The project uses the IBM HR Analytics Employee Attrition dataset, which contains 1,470 rows and 35 columns.

The project involves data preprocessing, data analysis, and visualization using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Plotly. An interactive Streamlit dashboard is also developed to present the analytical results in a clear and organized manner.

2.2 Project Objectives

- To understand and analyze the IBM HR Analytics Employee Attrition dataset.
- To perform data preprocessing and check the quality of the dataset.
- To apply Python Programming and Data Science techniques for data analysis.
- To identify patterns related to employee attrition.
- To present analytical results through charts and visualizations.
- To develop an interactive dashboard using Streamlit.

2.3 Project Scope

The scope of this project is to analyze employee-related data and identify patterns associated with employee attrition. It focuses on examining factors such as age, department, job role, monthly income, job satisfaction, overtime, and work-life balance.

The project includes data preprocessing, exploratory data analysis, and visualization using Python libraries. An interactive Streamlit dashboard is developed to present the analytical results in an organized manner. This project demonstrates the practical application of Python and Data Science in Human Resource Management and helps in understanding workforce trends.

CHAPTER 3

PROJECT REVIEW

3.1 Feasibility Study

The feasibility study determines whether the proposed Human Resource Analytics Employee Attrition using Python project can be developed successfully with the available resources. It considers the technical, operational, and economic aspects of the project.

3.2 Technical Feasibility

The project is technically feasible as it uses Python and commonly available Data Science tools. Pandas and NumPy are used for data processing, while Matplotlib, Seaborn, and Plotly are used for visualization. Streamlit is used to develop the interactive dashboard. The project can be implemented on a standard computer system.

3.3 Operational Feasibility

The project is operationally feasible because the Streamlit dashboard provides a simple and organized interface for exploring employee data. Its sections, including Dataset, Preprocessing, Visualization, and HR Insights, make the analytical results easier to understand and present..

3.4 Economic Feasibility

The project is economically feasible because it uses Python and its supporting libraries, which are freely available. It does not require expensive software or specialized hardware and can be developed using the available computer system and open-source tools.

3.5 Product Definition

The proposed product is an interactive Human Resource Analytics dashboard developed using Python and Streamlit. The system is designed to process, analyze, and visually present employee data in an organized manner. The project uses the IBM HR Analytics Employee Attrition dataset as input and performs data preprocessing and analysis before presenting the results. The Streamlit application provides different sections such as Home, Dataset, Preprocessing, Visualization, HR Insights, and About. These sections help users explore employee data and understand the analytical results in a clear and organized manner.

3.6 Problem Statement

Employee datasets contain a large amount of information related to age, department, job role, monthly income, job satisfaction, work-life balance, overtime, and employee attrition. Understanding this information directly from raw data can be difficult and time-consuming.

The project therefore focuses on developing a systematic solution for preprocessing, analyzing, and visualizing employee data. The analyzed information is presented through an interactive Streamlit dashboard so that important patterns related to employee attrition can be understood more easily.

3.7 Project Functionality

- The major functionalities of the proposed project are:
- Dataset Loading: Loading the IBM HR Analytics Employee Attrition dataset into the Python environment.
- Data Examination: Examining the structure and characteristics of the employee dataset.
- Data Preprocessing: Cleaning and preparing the data for further analysis.
- Data Analysis: Analyzing important employee attributes and patterns related to attrition.Data Visualization: Representing analytical results using appropriate charts and graphs.
- Interactive Dashboard: Presenting the dataset and analysis through a Streamlit-based interface.
- Organized Presentation: Providing separate sections for Home, Dataset, Preprocessing, Visualization, HR Insights, and About.

3.8 System Requirement

1. **Hardware Requirements:** Computer/Laptop, Minimum 4 GB RAM, Keyboard and Mouse ,Sufficient storage space for the dataset and project files
2. **Software Requirements** :

- Operating System: Windows
- Programming Language: Python
- Development Environment: Visual Studio Code
- Data Processing Library: Pandas
- Visualization Library: Plotly
- Dashboard Framework: Streamlit
- Dataset Format: CSV

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3.9 Methodology

The project follows a systematic data analysis methodology. The IBM HR Analytics Employee Attrition dataset containing 1,470 records and 35 columns is first loaded and examined to understand its structure and attributes. The dataset is checked for missing values and duplicate records before performing further analysis.

After preprocessing, the employee data is analyzed using Python libraries such as Pandas and NumPy. Different employee attributes, including Age, Department, Job Role, Monthly Income, Job Satisfaction, Overtime, and Years at Company, are examined. Suitable visualizations are created using Matplotlib, Seaborn, and Plotly. Finally, the analysis is presented through an interactive Streamlit dashboard.

Project Workflow

Raw Dataset → Data Examination → Data Preprocessing → Cleaned Dataset → Data Analysis → Data Visualization → Streamlit Dashboard → HR Insights

3.10 Acceptance Criteria

- The project will be considered successfully implemented when the following conditions are satisfied:
- The IBM HR Analytics Employee Attrition dataset is successfully loaded.
- The dataset containing 1,470 records and 35 columns is examined.
- Missing values and duplicate records are checked.
- The dataset is prepared for further analysis.
- Employee attributes related to attrition are analyzed.
- Appropriate charts and visualizations are generated.
- The Streamlit dashboard runs successfully.
- The Home, Dataset, Preprocessing, Visualization, HR Insights, and About sections are available.
- The analytical results are presented in a clear, organized, and interactive manner.

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CHAPTER 4

ANALYSIS.

4.1 Interface and Data External Flows

The project provides an interface through which the IBM HR Analytics Employee Attrition & Performance dataset can be accessed, processed, analyzed, and presented. The system takes the employee dataset as input and processes the available information before presenting the analysis through the Streamlit dashboard.

The major flow of information in the project is:

IBM HR Analytics Dataset → Data Preprocessing → Cleaned Dataset → Data Analysis → Data Visualization → Streamlit Dashboard → User

The dashboard provides different sections for accessing project information and analysis results. The user can navigate through the available sections and view the processed dataset and visual representations of the analysis.

4.1.1 ER-Diagram

Since this project is based on a CSV dataset rather than a relational database, a traditional database ER-diagram is not directly applicable. The IBM HR Analytics Employee Attrition & Performance dataset is treated as a structured collection of employee records. Each record contains information about an employee, including age, department, job role, monthly income, job satisfaction, and attrition. The analysis is performed using Python and Pandas.

4.2 Flowcharts

The project follows a sequential workflow from loading the raw dataset to displaying the final results.


HR ANALYTICS – PROJECT WORKFLOW
Employee Attrition & Performance
START
1. Load IBM HR Analytics Dataset (CSV)
2. Inspect Employee Records (1,470 rows • 35 columns)
3. Validate Data (Missing Values & Duplicates)
4. Preprocess Employee Data
5. Explore Employee Attributes (Age, Department, Job Role, Income, Satisfaction, Overtime)
6. Analyze Attrition Patterns
7. Create Visualizations (Matplotlib • Seaborn • Plotly)
8. Build Interactive Streamlit Dashboard
9. Present HR Insights & Results
END

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4.3 User Display and Report Format

The project uses an interactive **Streamlit dashboard** as the main user interface. The dashboard is organized into separate sections to make the information easy to access and understand.

The main sections are:

- **Home:** Provides an introduction and summary of the project.
- **Dataset Overview:** Displays important information about the dataset.
- **Pre-Processing:** Presents information related to data cleaning and preparation.
- **Visualization:** Displays the graphs and charts generated during the analysis.
- **About:** Provides information about the project.

The report presents the project development, methodology, analysis, visualizations, results, and conclusions in a systematic chapter-wise format.

4.4 Data Dictionary

The data dictionary describes the important attributes of the IBM HR Analytics Employee Attrition & Performance dataset. It helps in understanding the employee information used for analysis.

FIELD	DESCRIPTION
Age	Age of the employee.
Attrition	Indicates whether the employee has left the Organisation.
Department	Department in which the employee works.
JobRole	Job position of the employee.
MonthlyIncome	Monthly salary of the employee.
OverTime	Indicates whether the employee works overtime.
JobSatisfaction	Employee's job satisfaction level.
WorkLifeBalance	Employee's work-life balance rating.

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YearAtCompany	Number of years the employee has worked In the company.
PerformanceRating	Performance rating of the employee.

Note: Before final submission, the field names should be checked with the original IBM HR Analytics Employee Attrition & Performance dataset to ensure consistency between the report and dataset.

4.5 Functional and Performance Specifications

Functional Specifications

- The system should be able to:
- Load the IBM HR Analytics Employee Attrition & Performance dataset.
- Examine and preprocess the employee data.
- Check missing values and duplicate records.
- Analyze important employee attributes.
- Identify patterns related to employee attrition.
- Generate appropriate charts and visualizations.
- Display analytical results through the interactive Streamlit dashboard

Performance Specifications

The system should:

- Load the dataset correctly.
- Process the data without unnecessary errors.
- Display analysis results clearly.
- Generate visualizations correctly.
- Provide smooth navigation between dashboard sections.
- Present information in an organized and understandable manner

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CHAPTER 5

PROJECT PLAN

5.1 Development Schedule

The development of the **HR Analytics Employee Dasboard** project was carried out through a systematic and sequential process. The project was divided into different stages so that each activity could be completed and verified before proceeding to the next stage.

5.1.1 Project Development Stages

S.No.	Development Stage	Major Activity
1	**Project Planning**	Defining project objectives and requirements.
2	**Dataset Collection**	Collecting the HR dataset from Kaggle.
3	**Data Understanding**	Studying the structure, attributes, and contents of the raw dataset.
4	**Data Preprocessing**	Cleaning and preparing the dataset for further analysis.
5	**Data Analysis**	Analyzing important attributes and identifying patterns and trends.
6	**Data Visualization**	Creating suitable charts and graphs to represent the analysis results.
7	**Dashboard Development**	Developing the interactive Streamlit dashboard and organizing its sections.

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8	**Testing and Verification**	Checking the dashboard, data processing, and visualizations for proper functioning.
9	**Documentation**	Preparing the final project report and documenting the developed work.

he overall development process can be represented as:

Project Planning → Dataset Collection → Data Understanding → Preprocessing → Analysis → Visualization → Dashboard Development → Testing → Documentation

This sequential approach helped in maintaining an organized workflow throughout the development of the project.

5.2 Programming Language and Development Tools

The project was developed using Python and supporting Data Science tools. Each tool was selected according to its role in data processing, analysis, visualization, and dashboard development.
Writing

- **Python**: Used for loading the dataset, preprocessing, analysis, and implementing project logic.
- **Pandas:** Used for handling, cleaning, filtering, and analyzing employee data.
- **NumPy:** Used for numerical operations and data processing.
- **Matplotlib** and Seaborn: Used for creating charts and graphs to represent employee data.
- **Plotly**: Used for creating interactive visualizations to understand employee attrition patterns.
- **Streamlit:** Used to develop the interactive HR Analytics dashboard and display project results.
- **Visual Studio Code**: Used as the development environment for writing, editing, and managing Python code and project files**.**

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- **CSV Dataset:** The IBM HR Analytics Employee Attrition & Performance dataset is maintained in CSV (Comma-Separated Values) format. The dataset is examined and preprocessed, and the prepared data is used for analysis and visualization.
- **Development Environment:** The combination of Python, Pandas, NumPy, Plotly, Streamlit, and Visual Studio Code provides the required environment for implementing the project workflow, from data preprocessing and analysis to visualization and interactive dashboard development.

CHAPTER 6

DESIGN

6.1 Data Flow Diagram

6.2 Database Tables

The project uses employee data stored in CSV (Comma-Separated Values) format rather than a conventional relational database. Therefore, the dataset functions as the primary structured data source for the project.

The dataset contains 1,470 employee records and 35 columns. Each record represents an employee, while each column contains information related to employee attributes. The dataset is examined and preprocessed before being used for analysis and visualization.

Main Dataset Structure

Data Attribute	Description
Age	Represents the employee's age.
Attrition	Indicates whether the employee has left the Organization.
Department	Represents the employee's department.
JobRole	Represents the employee's job position.
MonthlyIncome	Represents the employee's monthly salary.
Over Time	Indicates whether the employee works overtime.
JobSatisfaction	Represents the employee's job satisfaction level.
WorkLifeBalance	Represents the employee's work-life balance Rating.
YearsAtCompany	Represents the number of years worked at the company.
PerformanceRating	Represents the employee's performance rating.

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The dataset is processed using **Pandas DataFrame** operations, and the cleaned data is subsequently used for analysis and visualization within the project.

CHAPTER 7

RESULTS AND DISCUSSION

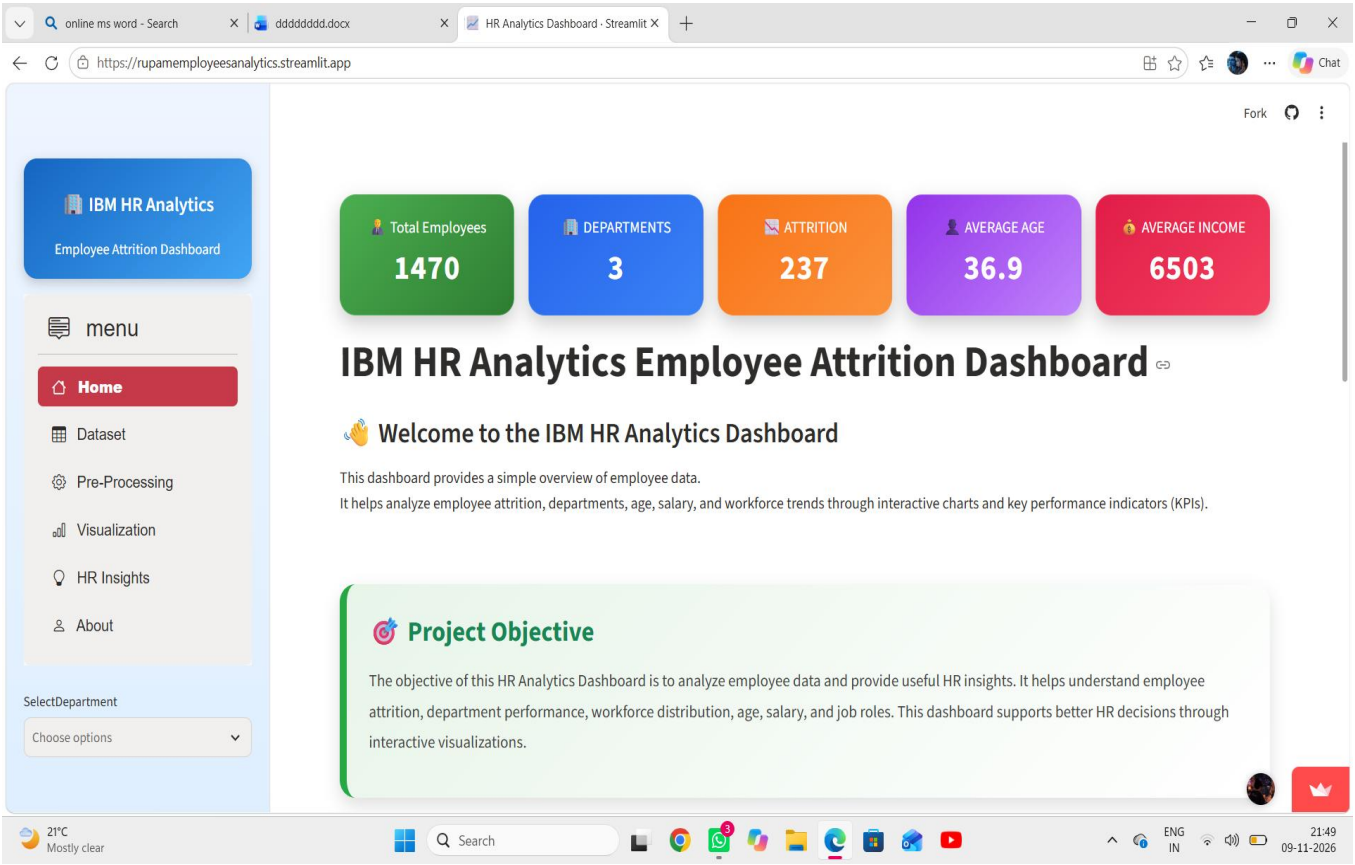
Screenshots with Description

The developed HR Analytics Employee Attrition project provides an interactive Streamlit dashboard for presenting employee data, preprocessing activities, analysis, and visualization results. The following screenshots demonstrate the major sections of the developed application.

7.1.1 Home Page

Figure 7.1: Home Page of HR Analytics Employees Attrition

The Home page serves as the introductory interface of the application. It provides an overview of the project and acts as the starting point for navigating through the different sections of the dashboard. The page provides access to the major components of the application through the navigation menu.

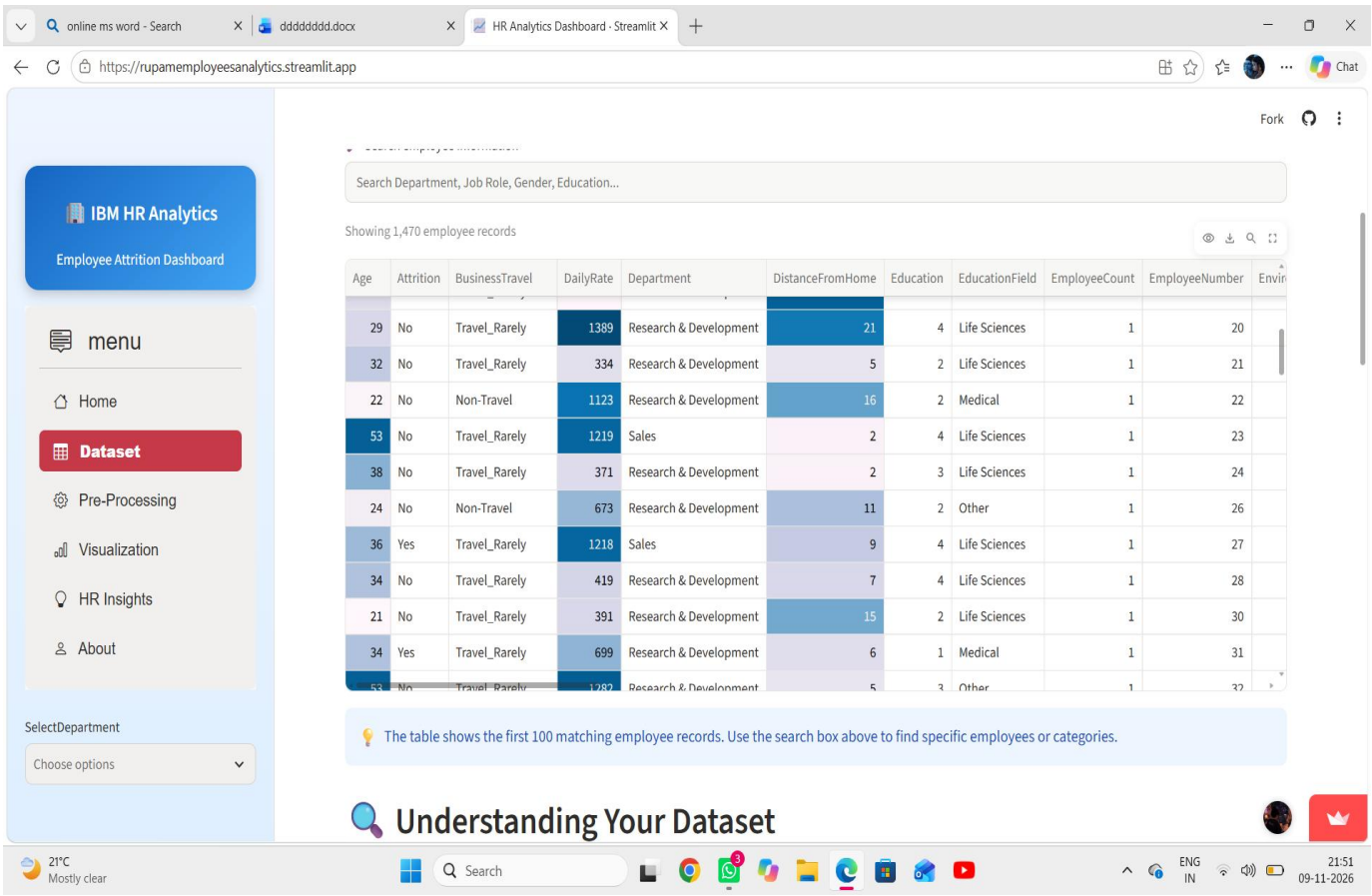


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7.1.2 Dataset

Figure 7.2: Dataset of HR Analytics Employees Attrition

The project uses the IBM HR Analytics Employee Attrition dataset, containing 1,470 employee records and 35 columns. It includes information about employee age, department, job role, income, satisfaction, performance, and attrition. The dataset is used to analyze employee-related patterns and understand the factors affecting employee retention.

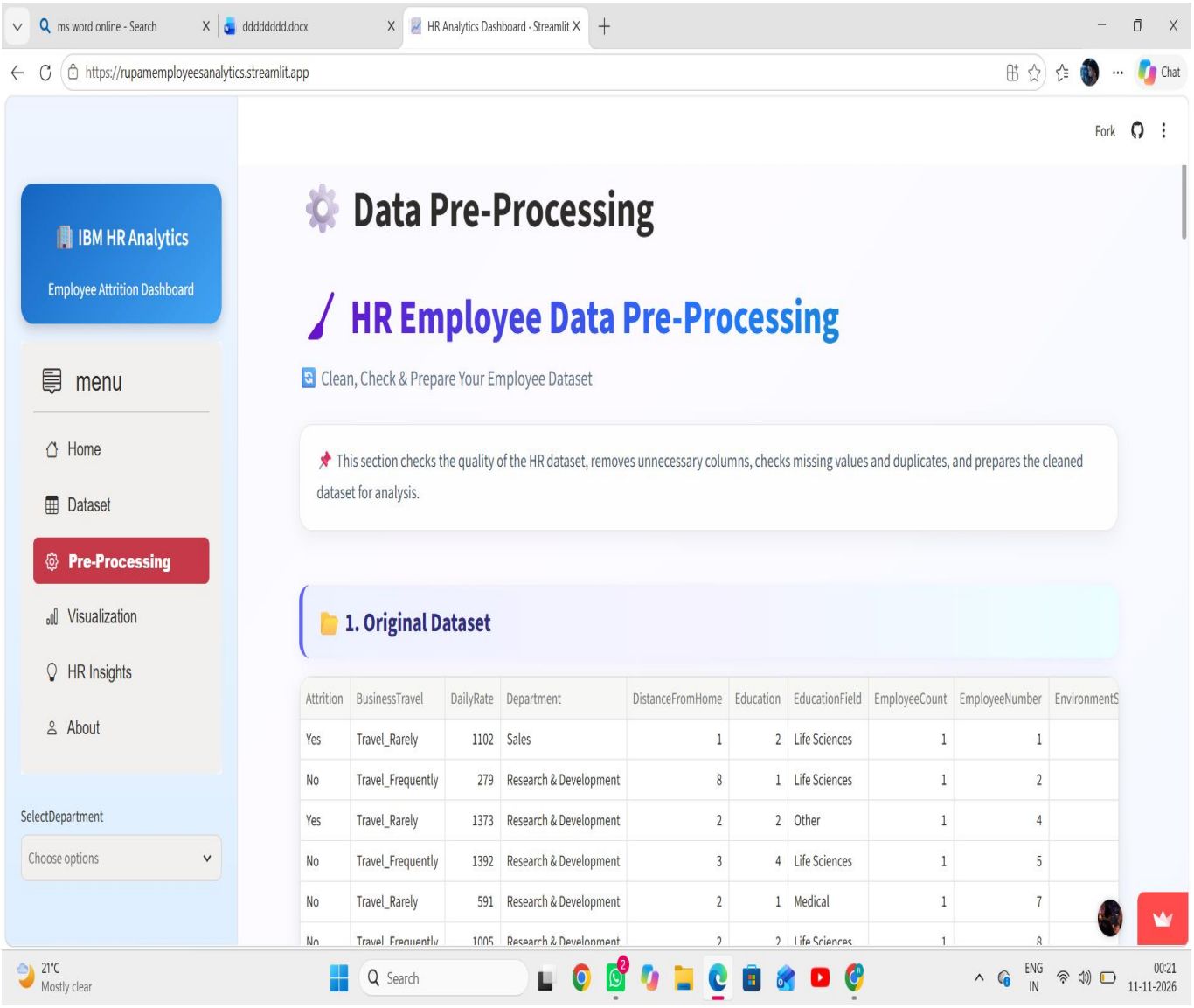


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7.1.3 Pre-Processing

Figure 7.3: Data Pre-Processing Page

The Preprocessing page displays the data cleaning and preparation steps performed on the HR Analytics dataset. It includes checking missing values, duplicate records, and data types. These steps help ensure data quality and prepare the dataset for further analysis and visualization.

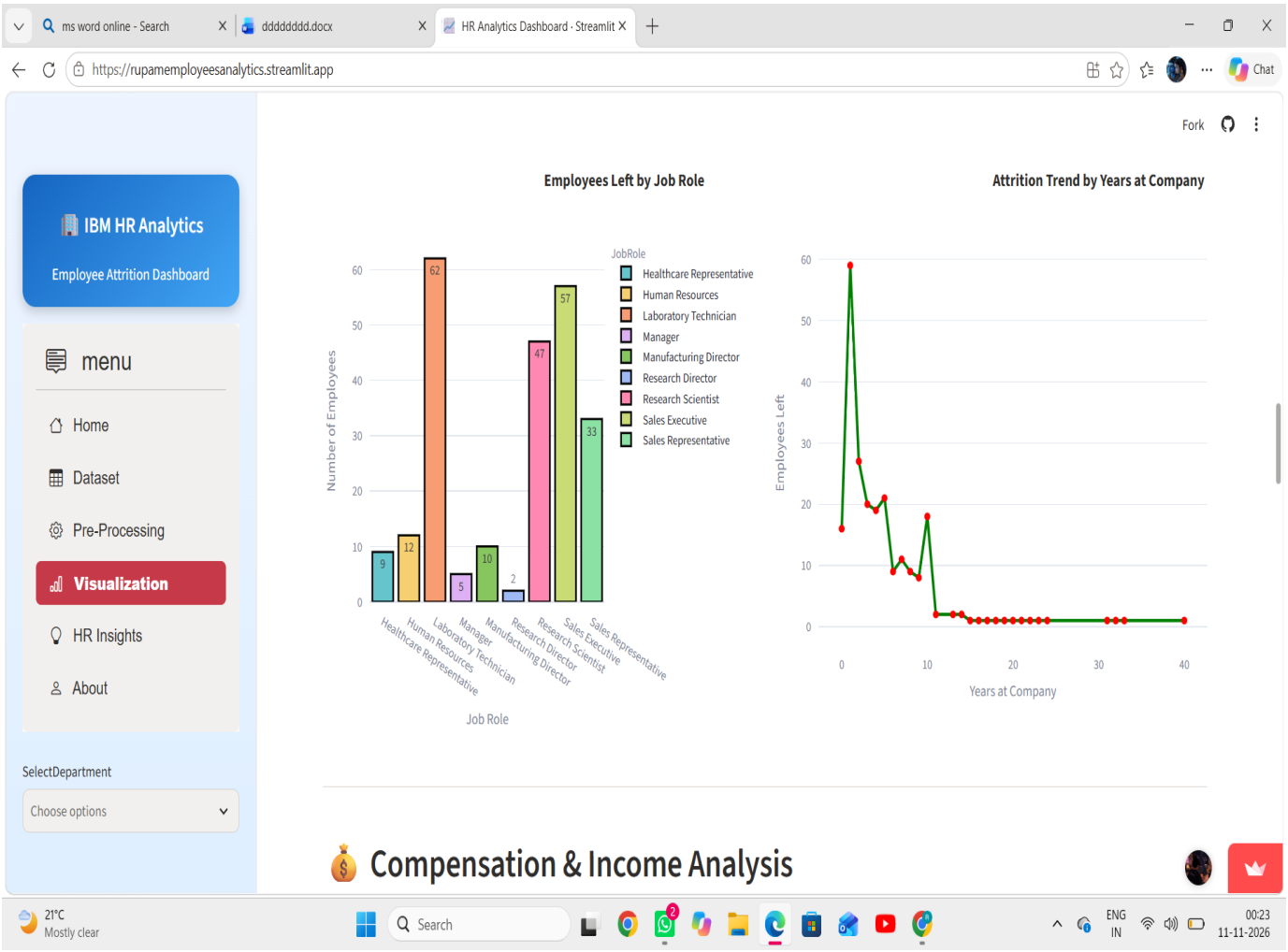


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7.1.4 Data Visualization

Figure 7.4: Data Visualization Page

The Data Visualization page presents employee data through interactive charts and graphs. It helps users understand attrition patterns based on age, department, job role, income, job satisfaction, and overtime. These visualizations make the analysis easier to understand and interpret.

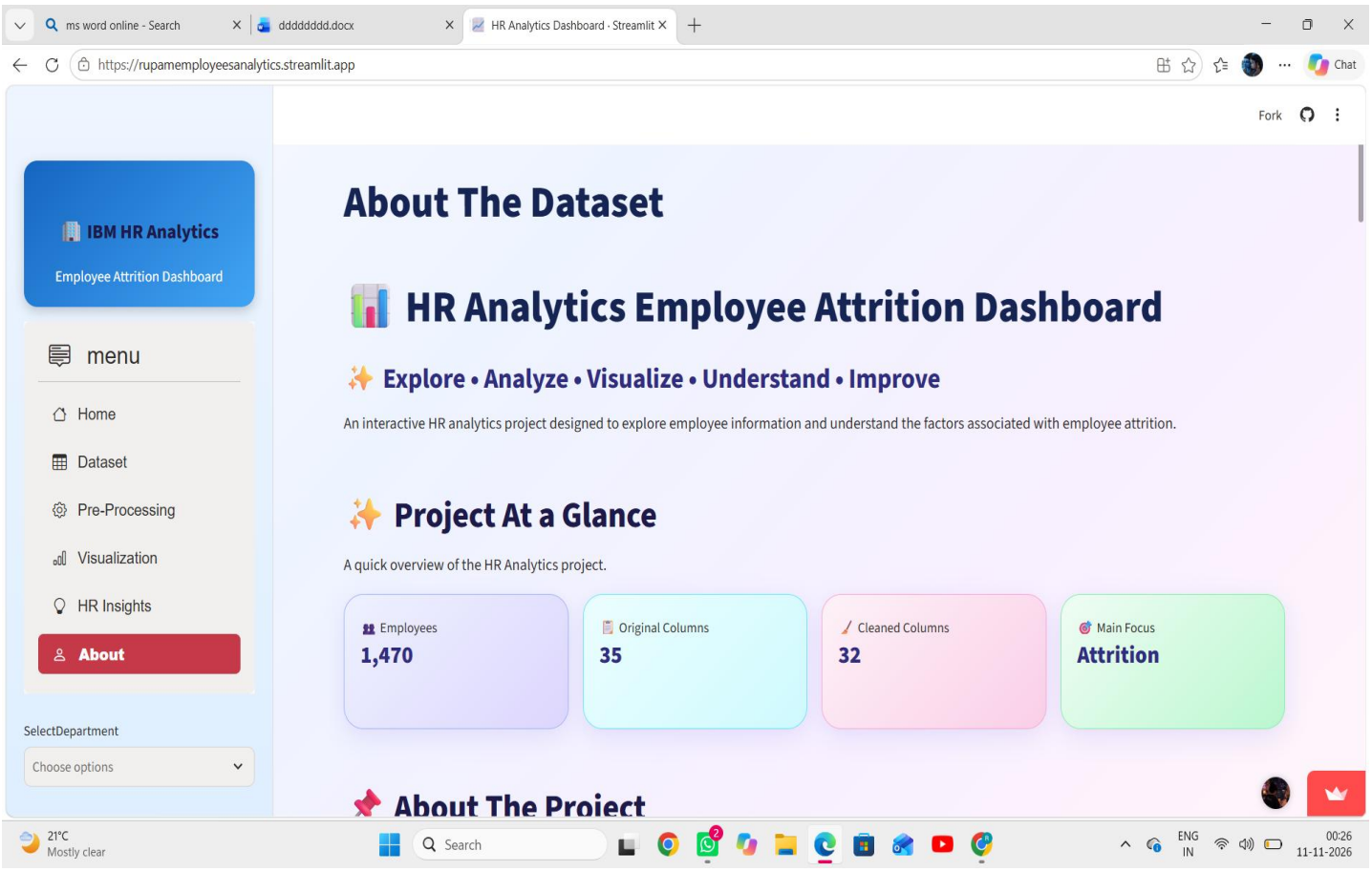


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7.1.5 About Page

Figure 7.5: About Page

The About page provides information about the HR Analytics Employee Attrition project, its objectives, tools, and technologies used. It explains how Python, Pandas, Plotly, and Streamlit are used to analyze employee data and present insights through an interactive dashboard**.**



7.2 Discussion of Results

The project successfully analyzes employee data to identify patterns related to attrition, job satisfaction, performance, and employee retention. The interactive visualizations help users understand employee-related trends and explore factors that may influence attrition. The Streamlit dashboard presents the analysis in a clear and user-friendly manner, making the results easier to understand.

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CHAPTER 8

CONCLUSIONS AND FUTURE SCOPE

8.1 Conclusion

The project “HR Analytics Employee Attrition using Python” successfully demonstrates the use of Python and Data Science techniques to analyze employee data. The project includes data preprocessing, analysis, and visualization to understand employee attrition patterns.

The Streamlit dashboard provides an interactive interface with sections such as Home, Dataset, Preprocessing, Visualization, HR Insights, and About. The project provided practical experience in Python programming, data analysis, and dashboard development. Overall, it helps transform employee data into meaningful insights.

8.2 Future Scope

The project can be enhanced by adding advanced filtering options and more interactive visualizations. Machine learning techniques can be introduced to predict employee attrition and identify factors affecting employee retention.

The dashboard can also be improved with updated employee datasets and additional analytical features to provide deeper insights and support better decision-making.

Overall, the project provides a foundation that can be further developed into a more comprehensive and interactive airline data analysis application.

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- Pandas Development Team. Pandas Documentation. Python Data Analysis Library.
- Plotly Technologies Inc. Plotly Python Documentation. Interactive Data Visualization Library.
- Streamlit Inc. Streamlit Documentation. Framework for developing interactive data applications.
- IBM HR Analytics Employee Attrition and Performance Dataset. Dataset used for data preprocessing, analysis, and visualization.
- Project Source Code. Python and Streamlit implementation developed for the HR Analytics Employee Attrition project.

- **ANNEXURE – I**

PROJECT SOURCE CODE

The project source code implements the HR Analytics Employee Attrition Dashboard using Python. It includes dataset loading, preprocessing, data analysis, visualizations, and the Streamlit interface. The dashboard contains Home, Dataset, Preprocessing, Visualization, HR Insights, and About sections.

ANNEXURE – II

DATASET AND ANALYSIS OUTPUT

The project uses the IBM HR Analytics Employee Attrition dataset, containing 1,470 employee records and 35 columns. The dataset includes employee information such as age, department, job role, income, satisfaction, performance, and attrition. The processed data is used for analysis and visualization, with results presented through the interactive Streamlit dashboard.

ANNEXURE – III

DASHBOARD AND VISUALIZATION OUTPUT

The developed HR Analytics Employee Attrition Dashboard presents employee data through interactive visualizations. The dashboard includes analysis of employee attrition, age, department, job role, monthly income, job satisfaction, performance, and overtime. The visualization results are presented through the Visualization section of the Streamlit application.